UNIT 10 · STANDARD 6.G.4

Surface Area Using Nets

Lesson 10-3

Name: ______ Date: ______ Class: ______

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box 6 × 4 × 2: SA = 2(24) + 2(12) + 2(8) = 88 in^2 — like wrapping paper needed

Surface area

The total area of all the flat sides of a solid.

Cut along the edges of a cereal box and unfold it flat — you see all 6 faces

Net

A flat shape that folds up into a solid.

A rectangular prism has 6 faces: top, bottom, front, back, left, right

Face

One flat side of a solid shape.

A rectangle drawn on paper is 2D — it has length and width but no depth

Two-dimensional

A flat shape with length and width, but no thickness.

A cube has 12 edges — the lines where two flat faces connect

Edge

The line where two flat sides of a solid meet.

Key Ideas & Notes

- Your class is designing custom wrapping paper for time capsule boxes.
- To figure out how much paper you need, you have to find the total surface area of each box.
- The trick?
- Unfold the box into a net so you can see every face laid flat!
- Identify each face of the rectangular time capsule box (l = 8 in, w = 5 in, h = 3 in). Find the area of each face pair, then calculate the total surface area.

Think About It

- How many faces does a rectangular prism have?
- What shapes do you see when the box is unfolded?
- Are all the faces the same size?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

To find how much wrapping paper a time capsule box needs, why is unfolding it into a net more helpful than looking at the 3D box?

Level 1 support

Start here: A net lays every face flat so you can see and measure all six faces.

A net helps because it shows all ____ faces laid ____.

Una red ayuda porque muestra las ____ caras puestas ____.

On the 3D box some faces are ____, but the net makes them all ____.

En la caja 3D algunas caras están ____, pero la red las hace todas ____.

Word bank: surface area | net | face | two-dimensional | edge

Listen for: *Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.*

Level 2

Is there more than one correct net for the same box? Justify your thinking.

- There ____ be more than one net because ____.
- Different nets still give the same surface area because ____.

EXPLORE

For the box l = 8, w = 5, h = 3, you found three face pairs. How did you use the net to organize the faces into pairs, and which pair had the largest area?

Level 1 support

Start here: Opposite faces match in size, so the six faces form three equal pairs.

The faces come in ___ pairs because opposite faces are ___.

Las caras vienen en ___ pares porque las caras opuestas son ___.

The largest pair is ___ × ___ = ___, counted twice for ___ in².

El par más grande es ___ × ___ = ___, contado dos veces para ___ in².

Word bank: surface area | net | face | two-dimensional | edge

Listen for: *Students explain opposite faces are congruent (3 pairs), identify the top/bottom 8×5 = 40 as largest (80 in² for the pair), and total to 158 in².*

Level 2

How can you tell which face pair will be largest just by looking at the dimensions, before doing any multiplication?

- The largest pair uses the ___ two dimensions because ___.
- I can predict the pair sizes by comparing ___.

CONNECT

A gift box is 14 × 10 × 6 in and one sheet of paper covers 400 in^2. Talk through how to find how many sheets are needed to cover the whole box.

Level 1 support

Start here: Find the surface area first, then divide by what one sheet covers and round up.

The surface area is ____ in^2 because SA = 2(14×10) + 2(14×6) + 2(10×6).

El área de superficie es ____ in^2 porque SA = 2(14×10) + 2(14×6) + 2(10×6).

I need ____ sheets because ____ ÷ 400 ≈ ____.

Necesito ____ hojas porque ____ ÷ 400 ≈ ____.

Word bank: surface area | net | face | two-dimensional | edge

Listen for: *Students compute SA = 280 + 168 + 120 = 688 in^2, then 688 ÷ 400 ≈ 1.72, and round up to 2 sheets to cover the whole box.*

Level 2

Why must you round the number of sheets up instead of down, and what does the leftover paper represent?

- I must round up because ____.
- The leftover paper is about ____ in^2 because ____.

REFLECT

A classmate found the surface area of a box but only multiplied two of the three face pairs by 2. How would you explain why every pair must be doubled?

Level 1 support

Start here: Each of the three face shapes appears twice on the box, so all three areas must be doubled.

Every face pair must be multiplied by ___ because each shape appears ___ times.
Cada par de caras debe multiplicarse por ___ porque cada forma aparece ___ veces.

If you forget to double one pair, the surface area will be too ___.
Si olvidas duplicar un par, el área de superficie será demasiado ___.

Word bank: **surface area** **net** **face** **two-dimensional** **edge**

Listen for: *Students explain a prism has opposite congruent faces (three pairs), so SA = 2lw + 2lh + 2wh, and missing a '2' undercounts one whole face.*

Level 2

How does looking at the net make it obvious that there should be six faces, not five? Generalize the rule.

- The net shows ___ faces because ___.
- In general, a rectangular prism always has ___ faces in ___ matching pairs.

Guided Examples

Example 1

A rectangular prism has l = 6 in, w = 4 in, h = 2 in. What is its surface area?

Solution: $SA = 2(6\times4) + 2(6\times2) + 2(4\times2) = 48 + 24 + 16 = 88$ in^2.

Answer: A. 88 in^2

Example 2

How many faces does a rectangular prism have?

Solution: A rectangular prism has 6 faces: top, bottom, front, back, left, and right.

Answer: A. 6

Example 3

Surface area is measured in which type of unit?

Solution: Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.

Answer: A. Square units (in^2, cm^2, ft^2)

Write About the Math

The Writing Revolution

I can explain my work using the words surface area, net, face, and two-dimensional.

1. Kernel Sentence

subject + verb

Model: Surface area is the total area of all the flat sides of a solid.
Área de superficie es el área total de todos los lados planos de un sólido.

Write a kernel sentence about surface area. Use a subject and a verb.
Escribe una oración base sobre área de superficie. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Surface area matters in math
Área de superficie importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Surface area matters in math because ____.
Área de superficie importa en matemáticas porque ____.

____

but *pero*

Surface area matters in math, but ____.
Área de superficie importa en matemáticas, pero ____.

____

so *entonces*

Surface area matters in math, so ____.
Área de superficie importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about surface area.
Di un hecho verdadero sobre surface area.

Surface area ____.

Question
Pregunta

Ask a question about surface area.
Haz una pregunta sobre surface area.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about surface area.
Muestra entusiasmo sobre surface area.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with surface area.
Dile a un compañero qué hacer con surface area.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

A net helps because it shows all ____ faces laid ____.
Una red ayuda porque muestra las ____ caras puestas ____.

On the 3D box some faces are ____, but the net makes them all ____.
En la caja 3D algunas caras están ____, pero la red las hace todas ____.

The faces come in ____ pairs because opposite faces are ____.
Las caras vienen en ____ pares porque las caras opuestas son ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Surface area is measured in which type of unit?

A. Square units (in^2, cm^2, ft^2)

B. Cubic units (in^3, cm^3, ft^3)

C. Linear units (in, cm, ft)

D. No units needed

Show your work:

2. A cube has edges of 5 cm. What is its surface area?

A. 150 cm^2

B. 125 cm^2

C. 150 cm^3

D. 30 cm^2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two boxes have the same volume of 60 in^3. Box A is $5 \times 4 \times 3$. Box B is $10 \times 6 \times 1$. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

Sentence starter: Box A: SA = 2(____) + 2(____) + 2(____) = ____ in^2. Box B: SA = 2(____) + 2(____) + 2(____) = ____ in^2. Box ____ uses less wrapping paper because ____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has l = 9 in, w = 5 in, h = 3 in. What is the surface area?

A. 174 in^2

B. 135 in^2

C. 174 in^3

D. 87 in^2

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Square units (in², cm², ft²) *— Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.*
2. **Try It 2:** A. 150 cm² *— A cube has 6 identical faces. Each face: $5 \times 5 = 25$ cm². Total SA = $6 \times 25 = 150$ cm².*
3. **Exit Ticket:** A. 174 in² *— SA = $2(9\times5) + 2(9\times3) + 2(5\times3) = 90 + 54 + 30 = 174$ in². Surface area uses square units (in²).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Surface area is the total area of all the flat sides of a solid.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).